Risk assessment and decision options as SysML-model vocabulary

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Scope and estimate. The halfarm design document holds a NIST SP 800-30 risk register (15 risks, 5-level likelihood and impact, the I-2 risk-level table, initial and residual values, mitigating requirements) outside the model. User decision 2026-10-07 18:06: the risk assessment becomes part of the integrated model, with an Ishikawa (cause–effect) view and decision diagrams as its views, inside the regular rendering workflow. ETA 4–5 hours. No benchmark.

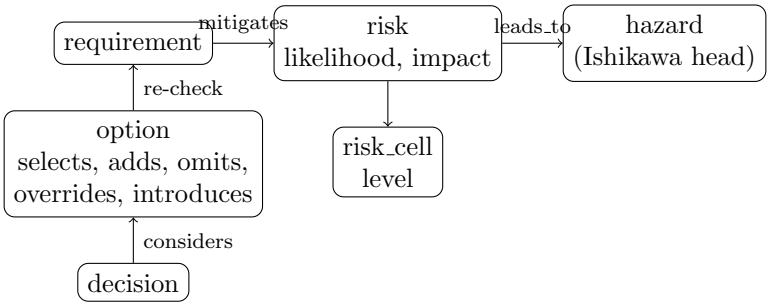
Semantics. New file `data/sysml/meta_sysml_risk.hymeko` (imports the structure vocabulary, which is not modified). Element kinds: `hazard` (an undesired top outcome, the head of an Ishikawa diagram), `risk` (a scenario with initial and residual likelihood and impact), `risk_category` (an Ishikawa bone), `risk_cell` (one cell of a likelihood × impact table: `likelihood`, `impact` → `level`), `decision`, `option`; scale `scale_level` (consumer levels with `rank`). Edges: `mitigates` (+ requirement…, − risk); `leads_to` (+ risk, − risk | hazard), a cause–effect chain; `considers` (+ decision, − option…); option effects `selects` (+ option, − candidate: replaces the line it is an alternative of), `adds` (+ option, − candidate, ∼ block it realizes), `omits` (+ option, − product), `overrides` (+ option, − constraint; edge fields `value`, `value_max`, `basis`), `introduces` (+ option, − constraint: checked only in that option). Levels are never stored on a risk: they come from the cells.

Ownership. Transformation: the vocabulary, a witness (`data/sysml/sysml_risk_witness.hymeko`) and a compile test; no engine change. Consumer (`02_design/sze/halfarm/model`): the register, cells, hazards, chains, decisions; views (register and matrix tables, Ishikawa SVG/TikZ, decision DOT) and checks in its `diagrams/`. The option evaluator reuses the consumer's requirement checker (scenario = selected set and constraint values changed, then re-checked).

Tests. Witness compiles and every kind/edge resolves; the transformation suite stays green. Consumer: the model reproduces the document register (levels and the summary 2/11/2 initial, 1/12/2 residual); model tests: complete 5×5 table, residual $\leq$ initial, acceptance rule, every risk reaches a hazard, a mitigating requirement refuted on the model is reported. Counter-tests: a removed cell, a residual above initial, a removed `leads_to`, a refuted mitigating requirement, a changed override value each produce exactly the expected finding.

Budget and risks. 15 risks, 25 cells, a few options: linear. Risk: hazard assignment and chains are a modelling judgement – tagged `<proposed>` for review. Risk: option assumptions (e.g. an enclosure humidity target) are stated on the override edge (`basis`) and printed with every verdict that depends on them.

State and rollback. `feat/material-lobes`, uncommitted changes of 2026-10-06/07; other workers' files untouched; no git writes from the sandbox. Rollback: delete the two new data files and the test module; the structure vocabulary is unchanged.


requirement
mitigates
risk
likelihood, impact
leads_to
hazard
(Ishikawa head)
re-check
option
selects, adds, omits,
overrides, introduces
risk_cell
level
considers
decision